Commute Distance and Residential Characteristics: Weathering the Storm

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Abstract Commuting distances in developing countries have received little attention, creating a significant gap in research. This study aims to address this crucial gap by utilizing the 76th NSSO round on "Drinking water, sanitation, and housing conditions"—a nationally representative dataset for our analysis. Research conducted in developed countries has shown that residential self-selection (RSS) has a negligible role in determining commute distance. In the context of a fast-developing country like India, this article investigates whether this phenomenon holds true. Our primary focus is on understanding the explanatory power of location and physical characteristics of households. Our analysis reveals that over two-thirds of job-related trips in India cover a maximum distance of less than 5 km, which holds true for both males and females. Additionally, when examining reasons for relocation, it becomes evident that relocation for employment-related reasons is rare, with residence being the core focal point. The association of these variables with commute distances indicates a negligible to weak relationship. Based on our results, we conclude that the inclusion of location and physical characteristics of households need not significantly contribute to the explanatory power of the model, and the FSM, or the four-step model, cannot be the sole basis for transport planning.

Keywords Household location · Residential characteristics · Commute distance

1 Introduction

The underlying premise in our examination is the belief that individuals tend to choose residences in close vicinity to their places of employment to mitigate the disutility associated with commuting [1, 17. This fundamental assumption forms the backdrop

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against which our analysis unfolds. We examine whether residential self-selection (RSS) influences commute distances in India, as it does in developed countries. Our analysis explores the impact of household location and physical characteristics on commuting patterns, aiming to provide insights for urban planning and policymaking.

We draw upon different strands in the literature, providing an overarching framework for the issues we address. Traditionally, transportation planning primarily centered around addressing long-term mobility requirements through the provision of sufficient transportation infrastructure. Over the past three decades, however, there has been a notable shift in focus. In this mobility-centric and supply-oriented planning approach, travel demand models played a key role in forecasting aggregate travel demand. These models were utilized to predict travel needs based on long-term socioeconomic scenarios and various alternative characteristics of transportation systems and land-use configurations.

Self-selected samples, identified as a significant error source in social science survey methods, can introduce bias, as seen in television program polls. Similarly, in land-use and travel studies, individual preferences for car travel may impact residential choices, influencing correlations between inner-city living and reduced car usage. These relationships may be more reflective of individual characteristics, such as income or attitudes toward car travel, than the location itself. Recognizing these nuances is crucial for understanding the dynamics between residential choices, travel behavior, and urban structure.

Research on travel behavior has extensively highlighted the influence of the built environment on travel patterns, as evident in various meta-studies such as those conducted by Ewing and Cervero in 2001 and 2010 [14, 15]. More recently, there has been a growing interest in the concept of residential self-selection (RSS), defined as the process wherein individuals choose residential locations that align with their travel attitudes and fulfill their preferences for commuting [2].

This renewed focus recognizes that the impact of the built environment on travel behavior is, to some extent, shaped by individual travel attitudes, which, in turn, influence the selection of specific residential areas [3]. For example, individuals with a preference for car-driving may be inclined to choose neighborhoods with ample parking facilities and comfortable driving conditions.

In the context of research correlating inner-city residential locations with lower car travel, it is suggested that such relationships may not be solely attributable to the location itself. Instead, individual characteristics of residents, such as income level or specific attitudes favoring reduced car usage or car ownership abstention, could be contributing factors [4].

A substantial body of literature has presented supportive evidence for the RSS hypothesis, as demonstrated in works by Cao et al. [2] Van Wee and [5]. However, it's noteworthy that some researchers have reported only moderate or negligible effects, as observed in the studies by Ettema and Nieuwenhui [6]. Additionally, there are discussions on the magnitude of RSS [3], and arguments have been put forth suggesting that criteria related to travel preferences may have minimal relevance in influencing residential location choices [6]. In addition, the literature on social science survey methods, self-selected samples are recognized as a significant source

of error, as highlighted by Heckman in [7]. An illustrative example involves television program hosts conducting polls among viewers of a debate program to gauge their opinions on a particular phenomenon. This method may introduce bias, as those who perceive the phenomenon as problematic might be more inclined to participate, potentially skewing the results.

In the context of India, a bulk of the studies which have analyzed commuting patterns have considered macro-level indicators [8–10]. We came across only a few studies which consider household characteristics for the analysis of commuting patterns [11, 12]. The former study examines commuting as rural–urban migration, while the latter research uses socio-economic household characteristics for investigating determinants of commuting patterns in women. This meant that the impact of physical characteristics and location were rarely considered for the analysis of commuting patterns. In contrast to this, transport and town planners heavily rely on the Four-Step Model (FSM), which overly emphasizes residential characteristics like density. It is observed that this overemphasis on residential characteristics may have led to the failure of public transport systems of the south. Our research is a modest attempt to bridge this gap, by understanding the role of residential characteristics on commuting distances.

After briefly analyzing the pertinent literature concerning commute patterns and residential self-selection and laying the basis for our investigation, the second section sets the objectives of the examination. The third section will provide detailed documentation on the data source and the methodology employed for our analysis. In the fourth section, we will discuss the inferences drawn from our findings. Lastly, the fifth section will delve into a comprehensive deliberation on the implications for policy formulation and their broader impact on society at large.

2 Objectives

The main goal of our research is to investigate how residential characteristics impact commute distance. To achieve this goal, we have divided our objective into three parts:

- Examine the influence of residential area characteristics on commute distance.
- Investigate the influence of household characteristics (physical) on commute distance.
- Determine the influence of physical infrastructure on commute characteristics.

3 Data Source and Methodology

Our analysis is based on the 76th round of the NSSO (National Sample Survey Organization) on Drinking Water, Housing Condition, and Sanitation. The NSSO is entrusted with the responsibility of collecting data on various important subjects

through their quinquennial rounds. The data collected follows a scientific methodology and provides a representative sample of India's population. Both the Central and State Governments rely on this data for the purpose of developing policies and making informed decisions.

In our research, we employ a purely quantitative methodology. Our focus is on the household as the unit of data, and our geographical scope encompasses the entire country of India. To gather information on the maximum distance traveled by males and females within a household, for accessing employment, we utilized the responses to questions 16.1 and 16.2 from the NSSO schedule 1.2.

The specific question posed was "What is the maximum distance normally traveled to the place of work?" This categorical variable, representing the maximum distance traveled, is divided into eight distinct buckets: "No travel," "less than 1 km," "1–5 km," "5–10 km," "10–15 km," "15–30 km," "greater than 30 km," and "not participating/member of specific gender not present in the household." As the data was collected separately for males and females, the analysis was conducted separately for each gender. This provided valuable insights and a deeper understanding of their travel patterns.

Our analysis commenced by examining the degree of similarity between the raw data (sample) and the weighted sample (~population). To accomplish this, we calculated the cosine similarity measure between the two datasets. Given the ordinal nature of the data, calculating a mean is inappropriate. To determine the degree of similarity between two data sequences, whose underlying generation process is unknown, cosine similarity offers a robust metric. Moreover, cosine similarity is a generalized form of correlation. This approach was deemed necessary due to the controversial nature of utilizing weights in regression analysis. This metric provides the basis for using the sample over the population in the regression analysis. Moving forward, we investigated the employment structure within households, specifically exploring whether the male(s) and/or female(s) of the household were engaged in economic activities, i.e., were a part of the labor force. For the present study, we define a commute trip as a journey undertaken for accessing employment. To ensure the focus remains on active commuters, households not actively participating in the labor market, such as those consisting solely of pensioners or relying solely on remittances, were excluded. The inclusion of this will not provide us with additional information as the commute distance category is "9," i.e., non-participation in labor market/"Specific Gender Not Present in the Household."

In the third step, we conducted an analysis of the reasons for relocation. This information was obtained from the responses to question number 4 in block 8, which inquired about the "Reason for movement to the Present Area." The question provided respondents with seven options to choose from. These options included: "Free/no rent," "Independent household," "Clean area," "Proximity to employment," "Other employment-related reasons," "Others," and "Not applicable." By examining these responses, our aim was to gain insights into whether employment served as the primary motivation for relocation.

Ensuring livable areas requires essential physical infrastructure, such as a reliable supply of drinking water, sanitation facilities, well-maintained roads, and electricity

[10]. With this in mind, we have categorized the variables into three distinct groups: area characteristics, physical characteristics of the residence including the nature of the ownership, and availability of infrastructure. The majority of the identified variables, both independent and dependent, fall into categorical categories. Consequently, we have conducted an analysis to determine the association between these variables.

The dependent variable in our study focuses on the maximum distance traveled by individuals of a household. It is important to note that the characteristics of this dependent variable align with the assumptions of multinomial logistic regression. Normalizing or scaling this variable would, however, diminish the interpretability of the results, as the baseline category serves as the reference point for all other categories. Furthermore, the dependent variable represents an observed outcome of an underlying binary decision-making process [12]. Alternatively, a latent variable model could be considered for analysis. However, the observed data, specifically the covariance matrix, exhibited low to negligible correlations among the variables proposed for the latent construct. This finding violates the underlying assumptions of both structural equation modeling (SEM) and partial least squares path modeling (PLSPM). Hence, we have opted to subject the data to stratified logistic regression, drawing inspiration from the works of Dittakavi and [12]. We assess the explanatory power of different variable categories, namely, area characteristics, physical characteristics of the household, and availability of infrastructure, by conducting separate regression analyses for each category. Following this, we construct an expanded regression equation. The equation can be represented as follows:

$$Y_i = a_i + b_i x_i + g_i z_i + d_i w_i + u_i \tag{1}$$

Here "Yi" is the maximum distance the male/female of the household commutes for accessing employment. "xi" denotes the area characteristics of the household. "zi" denotes the physical characteristics of the household, while "wi" denotes availability and nature of infrastructure and "ui" denotes the error term (refer Table 1).

The focus of the article is on the explanatory power and not the statistical relation of the variable(s). Hence, we compute the McFadden pseudoR2. We do acknowledge that there can be biases arising from the data collection process, but the study by Goel [13] using Commute data collected by Census-2011 shows similar commuting pattern, i.e., a distribution showing an exponential decay. Additionally, the data doesn't provide information on individual's commute distance and the commute mode. Thus, other aspects like the level of penetration of intermediate public transport and public transport in rural areas is kept outside the realm of the present research.

Table 1 Cosine similarity calculation between sample and weighted sample (~Population) based on employment status

	No work
No work	6203
Work	59890
Sum	66093

4 Results

4.1 Analysis of Gender-Based Participation in Economic Activity

In the preceding section, we mentioned that the household was identified as the unit of analysis. The incorporation of weights in regression analysis is a matter of contention and debate. Therefore, to determine whether the inclusion of sample weights can offer fresh insights, we calculate the similarity between the sample (raw data) and the weighted sample (~population) by computing the cosine similarity (see Tables 2 and 3 regarding the characteristics of both the sample and the population).

We observed a remarkably high cosine similarity of 0.99 between the raw data (sample) and the population (weighted sample). This indicates that operations on the sample will yield reliable and consistent results (Table 2).

Our research is centered around examining how residential characteristics affect the maximum distance individuals travel when commuting to access employment. As a result, we have excluded the records that represent homelessness, which amounts to 34 records or less than half a percent (0.04%) of the sample. With this understanding, we revisited the occupation structure of the households, which can be found in Table 3.

Based on our observations, it was discovered that slightly less than 6% (5.8%) of the households did not engage in any economic activity to support themselves. Instead, they relied on sources such as remittances, pensions, and other non-economic activities like begging and prostitution. On the other hand, in 34.04% of the households, both men and women actively participated in economic activities. An intriguing finding emerged, indicating that in over 50% (56.06%) of the households, only males were involved in economic activities, while in just over 4% (4.1%) of the households, females served as the primary earners. To align with our research objective, a deliberate decision was made to exclude households that did not partake in any form of economic activity.

Table 2 Employment cosine similarity between sample and population

	No work	Work	Sum
No work	18322781.00	11555626.00	29878407.00
Work	151610654.00	89616464.00	241227118.00
Sum	169933435	101172090	271105525

Table 3 Occupational structure of the household

	No participation/no women	Working
No participation/no men	5.8	4.1
Working	56.06	34.04

4.2 Maximum Distance Commuted for Accessing Employment

We collected data on the maximum distance traveled for accessing employment, specifically for males and females. Based on this, we conducted separate analyses for each gender. The sample size for males was 90.10%, which translates to 96,225 households. For females, the sample size was 38.14%, or 40,734 households. Figures 1 and 2 illustrate the analyses carried out on the maximum distance commuted by males and females, respectively.

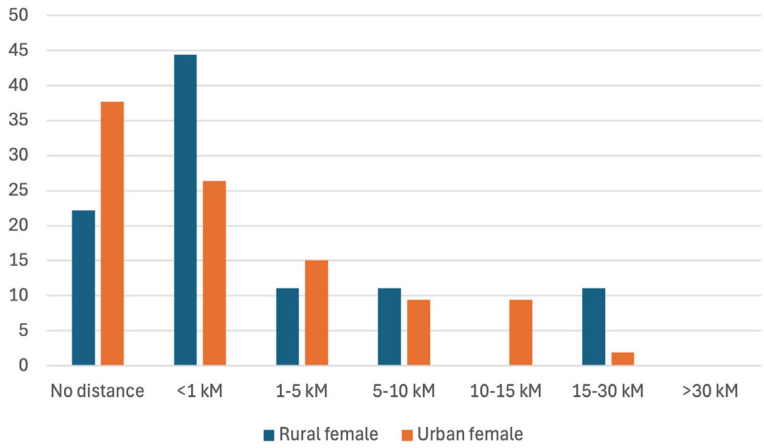

Fig. 1 Maximum distance traveled by females in urban and rural areas

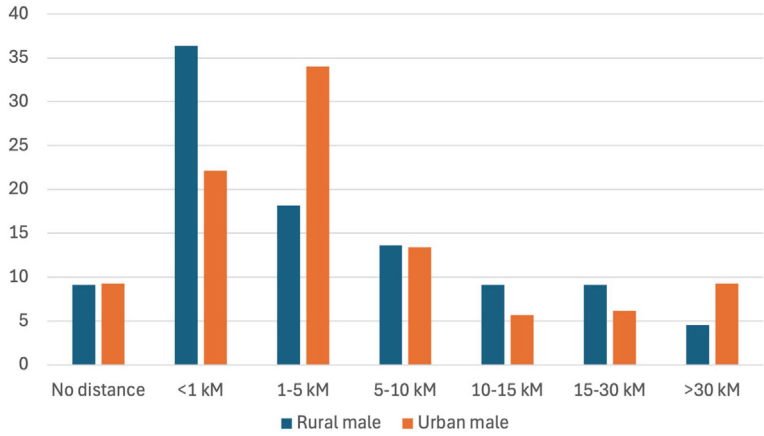

Fig. 2 Maximum distance traveled by males in urban and rural areas

The modal class of the maximum distance traveled for accessing employment differed between males and females. For males, the modal class was 1–5 km, while for females, it was reported as no distance. Consequently, the median class also varied: for males, the median class was "1–5 km," whereas for females, it was "less than 1 km." Based on a preliminary analysis, it can be inferred that the commute patterns of both males and females exhibit a long-tailed distribution. This inference is supported by research by Goel [13], which provides further evidence to corroborate our claims. We propose that individuals still demonstrate homing characteristics akin to animals [1] by minimizing the inconvenience of commuting, often by residing in close proximity to their place of work.

4.3 Residential Relocation and Employment

One of the factors which may lead to change in residence is attributable to employment. Using descriptive statistics, we examine the primary reasons which lead to change of residence. NSSO classifies residential shifts into six categories. These are "free/low rent," "independent accommodation," "accommodation in better locality," "employment-related reason: proximity to place of work," "other employment-related reason," and "others." In addition to the above categories, there is the category "not applicable," for households which have not changed their accommodation.

Upon examination, we observed that more than ninety-eight percent (98.31%) of households prefer to remain in their original residence. We attribute this reluctance to relocate to the homing instincts of human beings, as proposed by Marchetti [1]. Less than three-fourths of a percent (0.69%) of the total households relocate due to employment-related reasons. This includes both relocation due to proximity to employment and other employment-related issues. It is worth noting that employment opportunities are more abundant in urban areas. Therefore, we will further examine whether the rate of relocation is higher in urban areas compared to rural areas and investigate if there are any discernible patterns in these relocation patterns (Table 4).

The limited amount of relocation observed in rural areas can be attributed to the absence or minimal payment of rent for accommodation. As a result, a significant number of individuals, approximately 8.05 million according to Sharma [11], are crossing the rural–urban boundaries in search of employment opportunities. By comparing these findings to the maximum distance traveled by males and females, we can deduce that a substantial proportion of rural households, more than six-tenths, reside within a five kilometer radius of the urban fringe or the peri-urban region. This conclusion is supported by the works of Sharma [11] and Jain [10].

In contrast, approximately 1.57% of urban households relocate due to employment-related reasons, with 1.01% specifically relocating because of proximity to employment. In light of this information, we will now analyze the maximum

Table 4 Relocation patterns among urban and rural population

Reason	Free/low rent	Independent accommodation	Accommodation in better locality	Proximity to employment	Other employment-related reasons	Others	No relocation	Cosine similarity
Rural	0.03	0.12	0.06	0.08	0.05	0.13	99.54	0.99
Urban	0.35	0.4	0.58	1.01	0.56	0.7	96.4	

distance traveled by households that have relocated due to their proximity to employment. These findings have far-reaching implications, suggesting an inelastic residential market constrained by stringent floor space index regulations. Consequently, individuals are compelled to prioritize their current residences as primary living spaces. Moreover, these results must be interpreted in the context of the urban landscape, where over half of the population resides in informal settlements.

Table 5 shows the relocation patterns with respect to distances in rural and urban areas. It is evident that women living in urban areas have a clear preference for relocating to areas that are close to their workplace. This preference may be influenced by their extended family responsibilities, as they aim to balance work and family life effectively. On the other hand, in rural areas, a notable percentage of households (at least 11.11%) commute distances ranging from 15–30 km to access employment opportunities. These findings align with the research conducted by Sharma in [11], indicating a consistent pattern across different studies.

In the case of males, those residing in rural areas tend to prefer relocations within less than one kilometer, while males in urban areas tend to opt for a range of 1–5 km. This disparity can be attributed to the inadequate availability of transportation infrastructure and facilities in rural regions. The limited access to public transportation or commuting options likely influences the relocation choices of males in rural areas. Additionally, the rental gradient, which refers to the variation in rental prices across different locations, may also play a significant role in shaping these relocation patterns. Areas with lower rental costs may attract individuals who are willing to tolerate longer commuting distances. Due to social, occupational, and housing market rigidities, individuals may exhibit low mobility in response to disutility arising from commuting. Additionally, there may be a perpetuation of employment characteristics, as individuals are less exposed to alternative employment avenues and thus tend to pursue occupations like those they have been familiar with since childhood. These occupations may not necessarily be familial but rather prominent within their local area. At the aggregate level, these results suggest a multi-centric urban growth pattern. The extended rural commute for women can be attributed to the scarcity of non-agricultural employment opportunities in these regions. Furthermore, these relocations reinforce our previous assertion of an inelastic housing market, characterized by a limited supply of affordable housing options for low-income households.

Table 5 Maximum distance traveled by households that have relocated due to proximity to employment

	No distance	<1 km	1–5 km	5–10 km	10–15 km	15–30 km	> 30 km
Rural female	22.22	44.44	11.11	11.11	0	11.11	0
Rural male	9.09	36.36	18.18	13.64	9.09	9.09	4.55
Urban female	37.74	26.42	15.09	9.43	9.43	1.89	0
Urban male	9.28	22.16	34.02	13.4	5.67	6.19	9.28

4.4 Influence of Residential Area on Commute Pattern

According to Sharma and Chandrashekhar [11], approximately 8.05 million individuals transition from rural-to-urban areas in search of employment opportunities. Similarly, Jain [10] suggests that individuals living in rural regions have a higher travel requirement compared to their urban counterparts. However, it is worth noting that the former study does not provide specific data on the distances traveled, while the latter study's conclusions seem to be based on anecdotal evidence. In light of this context, the current section delves into an investigation of how area characteristics influence the maximum distance individuals travel when accessing employment opportunities.

Based on the analysis provided in Table 6, we can make several observations. Firstly, the rural/urban categorization demonstrates a weak association, while the area of residence exhibits a negligible association. Additionally, there is a stronger correlation between females and the rural/urban categorization compared to males. This suggests that females are more likely to encounter employment opportunities in urban areas. On the other hand, the relatively low association of males with the urban/rural categorization could be attributed to societal biases, wherein males are traditionally expected to be the primary breadwinners for their households. Consequently, the distance traveled may not play a significant role in their employment choices.

For the present analysis, we include monthly expenditure of the household as one of the explanatory variables. We include this as the cost of living is a function of the area characteristics.

Based on Table 7, we can conclude that the area characteristics of a residence, including the monthly expenditure of a household, can explain, at most, 4% of the commute relation for females. On the other hand, when it comes to males, the area characteristics of a residence can explain less than 2% of the variance. This finding supports our argument that, according to Indian social norms, male family members are expected to be the primary breadwinners. Furthermore, the 4% explanatory power only applies to distances exceeding ten kilometers. This result suggests that area characteristics and location might have an impact on the movement of females within these specific distance ranges.

Table 6 Maximum distance individuals travel when accessing employment opportunities

	Urban/rural	Area of residence
Male	0.15	0.03
Female	0.22	0.02

Table 7 Commute expenses as a percentage of the monthly expenditure

Distance	No distance	<1 km	1–5 km	5–10 km	10–15 km	>15 km	Not disclosed
Female	0.02	0.01	0.02	0.02	0.04	0.04	0.03
Male	<0.01	0.01	0.01	0.01	0.01	0.01	0.01

4.5 Influence of Physical Characteristics of Household on Commute Distance

In this section, we will delve into the examination of how the physical characteristics of a residence can influence the maximum commute distance. To gain a better understanding of the variables associated with the physical characteristics of the household, please refer to Table 8.

At first glance, it is evident that household characteristics have a greater influence on females compared to males. Among females, variables such as the condition of the household, type of kitchen, and availability of bathroom and toilet demonstrate a relatively strong association, while other variables exhibit a very weak association. Similarly, for males, the type of kitchen and availability of bathroom and toilet within the same premises also appear to play a significant role. Additionally, the utilization of the dwelling shows a relatively higher association in contrast to the remaining 11 variables, which exhibit a weak-to-negligible association. To administer the regression analysis in the present context, we include the number of rooms as one of the explanatory variables (Table 9).

After investigating the causal relationship, it is evident that household characteristics variables can explain a maximum of 9% of the variance among females and less than 7% of the variance among males. Specifically, among females, a higher pseudo-R^2 value is observed for distances exceeding 15 km, indicating that these variables have a stronger influence on commute distance in that scenario. On the other hand, for males, the high pseudo-R^2 value is observed for households where the male works from home, suggesting that these variables play a more significant role in shaping commute patterns for males engaged in home-based businesses.

This variation in the impact of household characteristics on females and males could be attributed to specific factors. For females, the need for security when traveling longer distances might be a prominent consideration, leading to a stronger association between household characteristics and commute distance. Conversely, for males, the emphasis on comfort and convenience in operating a home-based business could drive a higher correlation between household characteristics and commute behavior. Overall, these findings highlight the nuanced relationship between household characteristics, gender, and commute distance, underscoring the significance of context-specific factors in understanding commuting patterns.

Table 8 Influence of physical characteristics on residence

	Tenure of dwelling	Type of dwelling	Use of dwelling	Condition of dwelling	Availability of domestic electricity	Kitchen type	Bath and toilet together	Availability of drinking water	Availability of sufficient water other than drinking	Ventilation	Problem of vector (s)	Human feces	Stagnant water	Instance of flood
Female	0.05	0.1	0.09	0.18	0.04	0.15	0.15	0.06	0.07	0.09	0.06	0.04	0.05	0.03
Male	0.02	0.09	0.13	0.07	0.04	0.1	0.1	0.02	0.02	0.06	0.04	0.03	0.02	0.02

Table 9 Pseudo-R^2 values for association between household characteristics and commuting distance by gender

	No distance	<1 km	1–5 km	5–10 km	10–15 km	>15 km	Not disclosed
Female	0.04	0.02	0.03	0.03	0.05	0.08	0.06
Male	0.06	0.01	0.01	0.01	0.02	0.02	0.02

4.6 Influence of Physical Infrastructure on Maximum Distance Commuted

Built infrastructure, such as roads and access to water, as well as the provision of infrastructural services like waste disposal and vector control, are essential factors in creating livable spaces. They greatly contribute to the overall quality of life. Moreover, these improvements in infrastructure lead to increased accessibility and better health outcomes, resulting in women being able to actively engage in economic activities. With this understanding in mind, we conduct an analysis to compute the Cramer's V statistic. This statistic helps us measure the level of association between the maximum distance commuted and various infrastructural attributes (Table 10).

Except for the primary source of drinking water, which has a weak association of 0.1 with females, all the other variables show very weak or negligible associations. Similarly, when considering males, all the infrastructural variables exhibit a very weak association. Based on these results, it can be inferred that infrastructure and infrastructural services may not have a significant impact on the maximum distance commuted for accessing employment.

To examine causality, we conducted a logit regression analysis using these variables. Our hypothesis is that as per capita incomes increase, individuals tend to prefer residential areas that offer basic infrastructure and infrastructural services. Additionally, rather than solely relying on the functional division of livable space, individuals tend to prioritize larger floor areas. Therefore, we included usable floor space as a continuous variable in our analysis. Upon subjecting our results to a variance inflation factor (VIF) test, we discovered a high correlation between the primary source of drinking water and the primary source of water excluding drinking water. As a result, we decided to exclude the variable depicting the primary source of water from our analysis (Table 11).

Table 10 Cramer's V statistic for association between maximum commuting distance and infrastructural attributes by gender

	Vector control	Garbage collection	Road	Principle source of drinking water	Supplementary source of drinking water	Principle source of other water	Type of drainage
Female	0.07	0.08	0.07	0.1	0.06	0.09	0.09
Male	0.05	0.06	0.06	0.06	0.05	0.06	0.07

Table 11 Pseudo-R^2 values from logit regression analysis for association between primary source of water and commuting distance by gender

	No Distance	<1 km	1–5 km	5–10 km	10–15 km	>15 km	Not disclosed
Female	0.05	0.02	0.03	0.05	0.07	0.08	0.07
Male	0.01	0.02	0.01	0.01	0.02	0.02	0.02

The regression results provide further evidence to support our previous claim that infrastructure and infrastructural services have a greater impact on females compared to males. Among females, the infrastructural variables accounted for a maximum of 8% of the variance, whereas for males, the infrastructural attributes explained less than 3% of the variance. These findings suggest that to increase female work participation, it is crucial to improve the quality of infrastructure and infrastructural services. By enhancing the infrastructure, such as roads, water access, waste disposal, vector control, and other services, we can create an environment that empowers and enables women to participate more actively in the workforce.

4.7 The Extended Model

In this section, we focus on analyzing the explanatory power of the extended regression model. Specifically, we combine all the variables related to the three categories: area, physical condition of the house, and infrastructure. By merging these variables, we aim to gain a comprehensive understanding of their collective impact. Our findings indicate that the overall explanatory power of the regression model is highest at 11% for females and 7% for males (please refer to Table 12 for detailed information). It is worth noting that, similar to our earlier regressions, the results vary across different distance categories. This suggests that the influence of these variables on the outcome is not consistent throughout the entire range of distances.

The variables examined in our analysis demonstrate a stronger influence on women compared to men. This finding reinforces our presumption that Indian society tends to perceive males as the primary breadwinners for their families, as indicated by the low R-squared value. Interestingly, for men, the highest R-squared value is observed when there is "no distance" involved, suggesting that the variables play a significant role when there is no commuting required. On the other hand, for women, the highest R-squared value is observed when the commuting distance

Table 12 R-squared values of the extended regression model for females and males across different commuting distance categories

	No distance	<1 km	1–5 km	5–10 km	10–15 km	>15 km	Not disclosed
Female	0.08	0.03	0.06	0.06	0.08	0.11	0.09
Male	0.07	0.02	0.02	0.02	0.03	0.03	0.03

exceeds 15 km. These results align with those of the preceding sections, further supporting the conclusion that improving the quality of dwelling, infrastructure, and infrastructural services is likely to enhance female workforce participation. While these observations are anecdotal, they suggest that the increased work participation in urban areas may be attributed to the presence of better infrastructure compared to rural areas, rather than solely improvements in lifestyle conditions.

5 Conclusion

We find that despite differences in the average and median commute distances between men and women, there is a significant level of similarity in their commuting patterns. This finding contradicts Jain's [10] conclusions, which suggested that men commute more than women. When examining maximum distances traveled for employment, we observed that over 70% of households travel less than 10 km for work. This indicates that most of the 12.05 million individuals crossing the rural–urban boundary for work live near urban settlements, suggesting rural–urban migration may not be as significant as previously assumed.

While numerical differences in relocation patterns between rural and urban areas were observed, these differences were not statistically significant. Over 95% of sampled households continue to live in their original areas, with only about 1% relocating due to employment. This highlights a preference for finding employment close to home rather than relocating.

In developing economies, urban landscapes have evolved organically, intertwining commercial and residential areas. This has led to distances that are easily accessible using sustainable modes of transportation like walking and cycling. Planners should facilitate this organic growth rather than focusing solely on central business districts or industrial hubs based on central place theory. Our research sheds light on commuting patterns in rapidly developing countries like India, emphasizing the preference for shorter commutes and the limited role of physical location and infrastructure characteristics in influencing commuting behavior. These findings have critical implications for urban planning and Sustainable Development Goals (SDGs).

Shorter commutes reduce travel time, congestion, and vehicle emissions, directly improving air quality and supporting SDG 3 by decreasing pollution-related illnesses. By promoting shorter commutes and reducing reliance on automobiles, carbon footprints can be significantly lowered, aligning with SDG 13 on climate action. Sustainable transportation infrastructure, as emphasized in SDG 9, enhances accessibility, mobility, and social inclusion, while urbanization aligned with SDG 11 fosters liveable cities. Policymakers can leverage these insights to encourage sustainable commuting and improve urban quality of life.

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